Higher National Diploma in Quantity Surveying

2nd Year, Second Semester Examination – 2020

QS22073 - Structural Design & Detailing

Instructions for Candidates:

Answer 05 Questions only

Make reasonable assumptions wherever necessary

No. of questions :06

No. of pages : 05

Time :03 hours

Q1 Design reinforcements for a span and a support where Maximum bending moment occurs in the following continuous one-way slab arrangement, assuming a cover of 25mm and other dimensions are as shown below:

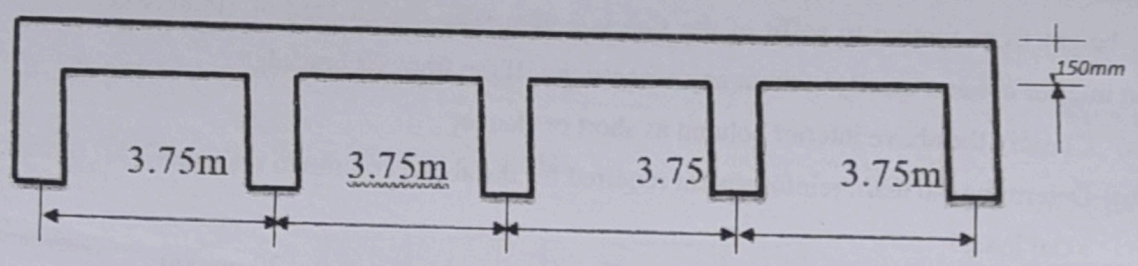


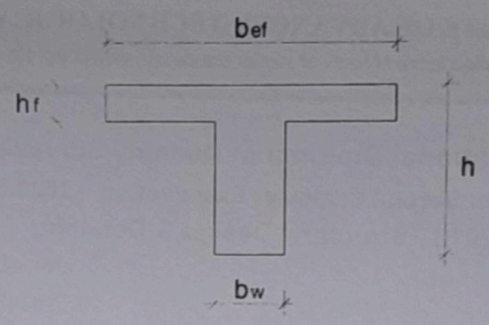
Fig (1)

- Imposed load, $Q_k = 4\ kN/m^2$
- Finishes of slab = $1.5\ kN/m^2$
- Density of Concrete = $24kN/m^3$
- The characteristic material strengths of concrete and reinforcements are 30N/mm2 and 460N/mm2 respectively.

(Total 20 marks)

Q2 A flanged beam shown in fig (2) is subjected to a sagging moment of 175kNm. Width of the flange (b_{ef}) is 600mm and width of the web (b_w) is 250mm. Thickness of the flange (h_f) is 150mm and overall height (h) of the section is 575mm. Design main reinforcements by assuming ,

- Cover to the main reinforcements = 25mm
- Characteristic strengths of materials are f_{cu} =30 N/mm^2 and f_y = 460 N/mm^2



Fig(2)

(Total 20 marks)

Q3 The reinforced concrete columns of size of 300 X 300 mm are used to support a 150mm thick roof slab. The column is connected to the footing at the bottom and to the slab at the top. The column height from footing to soffit of the slab is 3.00m. The design load of 1000kN is applied to an interior column axially without any eccentricity. If the frame is braced,

a) Classify the above interior column as short or slender
b) Determine the main reinforcement required for the above column to resist the axial load.

f_{cu} = 25 N/mm^2 , f_y = 460 N/mm^2 (Total 20 marks)

Q4 A simply supported reinforced concrete rectangular beam is used to carry a dead load of 12.5 kN/m (including self weight of the beam) and imposed load of 11kN/m.

Following Details are given

- No redistribution of load
- Characteristics strengths of materials are f_{cu} =25 N/mm^2 and f_y = 460 N/mm^2
- Cover to the stirrups = 25mm
- Width of the support = 300mm
- Clear spanning between support = 6.0m
- Size of the beam = 300 X 450 mm

Design the bending reinforcements of above beam and do the relevant checks.

(Total 20 Marks)

Q5 A timber beam that belongs to D70 is used to withstand a medium term uniformly distributed load of 1.5kN/m under service class 3 conditions. The beam is simply supported at the ends without overhanging of each ends. The beam is 75 x 150 mm and clear span 4.5m. Notch equal to half of the depth of cut at the support for better sheeting and the bearing width is 75mm.

a) Check for bending

b) Check for shear

(Total 20 marks)

Q6 Two S275 steel plates form a lap joint as shown in fig (3) with four 20 mm diameter Grade 8.8 steel black bolts. Calculate,

a) The maximum load that the joint can support

b) The minimum lap length that have to be provided

Clearly mention the assumptions that you made.

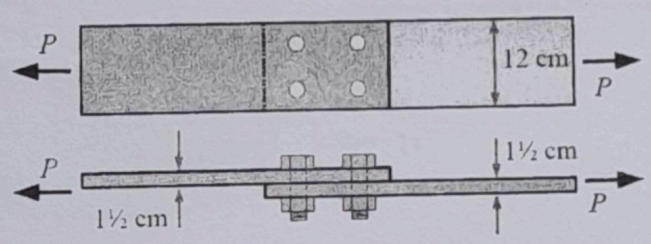


Fig (3)

(Total 20 marks)

Area's of Reinforcement
Cross-sectional areas of groups of bars (mm^2)

Bar Size	Number of Bars * Denotes non-preferred sizes									
	1	2	3	4	5	6	7	8	9	10
6mm	28.3	56.5	84.8	113	141	170	198	226	255	283
8mm	50.3	100	151	201	251	302	352	402	452	503
10mm	78.5	157	236	314	339	471	550	628	707	785
12mm	113.1	226	339	452	566	679	792	905	1718	1131
16mm	201.1	402	603	804	1005	1206	1407	1608	1810	2011
20mm	314.2	628	942	1257	1571	1885	2199	2513	2827	3142
25mm	490.9	982	1473	1963	2454	2945	3436	3927	4418	4909
32mm	804.2	1608	2413	3217	4021	4825	5630	6434	7238	8042
40mm	1257.0	2513	3770	5026	6283	7540	8790	10053	11310	12566
50mm*	1963.5	.3927	5890	7854	9812	11781	13744	15708	17671	19635

Cross-sectional areas for Specific Spacings in mm^2 / Metre Width

Bar Size	Spacing of Bars (Millimetres) * Denotes non-preferred sizes									
	75	100	125	150	175	200	225	250	275	300
6mm *	377	283	226	188	162	141	125	113	103	94
8mm	670	503	402	335	287	251	223	201	183	168
10mm	1047	786	628	524	449	393	348	314	285	262
12mm	1508	1131	905	754	646	565	503	452	411	377
16mm	2681	2011	1608	1340	1149	1005	894	801	731	670
20mm	4189	3142	2513	2094	1795	1571	1396	1257	1143	1047
25mm	6545	4909	3927	3272	2805	2454	2182	1963	1785	1636
32mm	-	8042	6434	5362	4596	4021	3574	3217	2924	2681
40mm	-	-	10053	8378	7181	6283	5587	5027	4571	4189
50mm*	-	-	-	13090	11220	9617	8727	7854	7140	6545

Shear strength of bolts

Bolt Grade	Shear strength , p_s ($N/mm^{2)}$
4.6	160
8.8	375
10.9	400

T_{16} @ 175 mm

Bearing Strength of Bolts

Bolt Grade	Bearing Strength
4.6	460
8.8	1000
10.9	1300

Bearing Strength of Connected plates

Steel Grade	S275	S355	S460
Bearing Strength(N/mm^2)	460	550	670

Bearing Capacity of bolt, $P_{bb} = dt_b p_{bb}$

Bearing capacity of connected plates, $P_{bs} = k_{bs} d t_p p_{bs}$ but $P_{bs} \leq 0.5 k_{bs} e t_p p_{bs}$

k_{bs} = 1 for Standard clearance holes

e = End distance

d = Nominal diameter of bolt

t_p = Thickness of the connected plates

Tensile stress area

Nominal size and tread diameter (mm)	Tensile stress area (mm^2)
15	84.5
16	157
20	245
22	303
24	353